

KOMAL SHAHID

AI/ML Engineer • GenAI & RAG Systems • Forward Deployed Engineering

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SUMMARY

AI/ML Engineer with 4 years of production machine learning in regulated financial services at Discover (now Capital One). Built a production RAG pipeline from scratch for enterprise GenAI use cases and work across the full GenAI lifecycle, from ingestion and embedding through retrieval, evaluation, and real-time inference. Research-grade thinking, production-bound: I take AI systems from prototype to production alongside the teams who depend on them.

EXPERIENCE

Discover Financial Services (now Capital One) Chicago, IL
AI/ML Engineer May 2024 – Present

- **Production RAG pipeline:** built a Retrieval-Augmented Generation pipeline from scratch to serve enterprise GenAI use cases.
- Delivered across the full GenAI lifecycle as part of the engineering team: document ingestion, embedding, retrieval, evaluation, and real-time inference.
- Architected the pipeline to scale across multiple business use cases instead of a single application.
- Deployed end-to-end AI models with the ML Engineering team, with hands-on AWS deployment work.

Application Engineer, Machine Learning Engineering Jun 2022 – May 2024

- Optimized the end-to-end model deployment pipeline for the Machine Learning Engineering team.
- Designed and built a data quality application using OOP principles and cloud data; it runs automated data quality checks and gives data scientists detailed logging during model development and deployment.
- Automated model performance tracking in Python, SQL, and SAS, cutting monitoring time from **4 hours to under 10 minutes**.
- Analyzed rising overlimit fraud cases in SQL, helping the fraud team develop and implement loss-mitigation strategies.

SELECTED PROJECTS

Hybrid RAG Pipeline • independent LangChain • ChromaDB • BGE • BM25
• Combined BM25 and dense BGE retrieval with Reciprocal Rank Fusion, cross-encoder reranking, and HyDE query expansion to fix the exact-term failures of dense-only search; answers cite their source passages.

LoRA Fine-Tuning, GPT-2 Quote Tagger • independent PEFT • Hugging Face
• Trained **590K of 125M parameters (0.47%)** with rank-16 adapters, a >99% memory reduction, fully on CPU with no GPU.

AI-Driven Content Strategy Engine • MS capstone OLS • Gradient Boosting • LLM
• Modeled engagement across 12,000 posts; LLM drafts conditioned on the winning patterns showed a **+37% predicted engagement lift (95% CI 15–64%)**.

Mental Health Treatment-Seeking Classifier • MS capstone scikit-learn • XGBoost • SHAP
• Compared logistic regression, random forest, XGBoost, and a neural network; picked logistic regression (**AUC 0.723**) for interpretability, with SHAP explanations and a demographic fairness audit.

EDUCATION

Bellevue University — Master of Science, Data Science June 2026
DePaul University — Bachelor of Science, Mathematics and Computer Science Aug 2020

TECHNICAL SKILLS

GenAI & ML: RAG, hybrid retrieval, reranking, embeddings, LLM evaluation, LoRA/PEFT, LangChain, ChromaDB, Hugging Face, scikit-learn, XGBoost, SHAP

Languages: Python, SQL, R, SAS, C++, Java, Rust

Cloud & Platforms: AWS (hands-on, including Amazon SageMaker), Docker, CI/CD, Snowflake, GitHub, Linux, Tableau